Proactive Aviation Anomaly Detection: Real-Time Checkpointing and Explainable AI

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***Abstract*—Rapid advancements in machine learning have increasingly been applied to aviation safety, primarily focusing on reactive anomaly detection and post-incident analysis. However, the practical integration of these advanced models into real-world, real-time aviation operations remains significantly underexplored. This paper introduces a dual-contribution framework targeting the proactive prediction of aviation safety incidents, specifically high-speed exceedances. First, we establish a mathematically rigorous real-time checkpointing framework that evaluates model predictions at 25%, 50%, 75%, 90%, and 100% of flight sequence completion, simulating instantaneous decision-making during approach and landing. We benchmark a comprehensive tier of models, ranging from aviation-specific baselines (Base GRU, MHCNN-RNN) to cross-domain state-of-the-art architectures (Inception Time, Gated Transformer Networks) and adaptive early-classification systems (EARLIEST). Second, we conduct an extensive explainability audit utilizing Grad-CAM for temporal concept formation, Monte Carlo Dropout for epistemic uncertainty quantification, and perturbation-based methods (LIME) for rigorous spatial diagnosis. By bridging the gap between predictive accuracy and safety-critical trustworthiness, this research demonstrates how black-box sequence models can be transformed into verifiable, proactive alert systems suitable for human-in-the-loop aviation deployment.**

***Index Terms*—Aviation safety, anomaly detection, early classification, deep learning, explainable AI, uncertainty quantification**

I. Introduction

The escalating volume of air travel necessitates a concurrent emphasis on proactive aviation safety protocols [1]. The confluence of rapid advancements in aviation technology, coupled with mounting operational pressures, has unfortunately contributed to a rise in aviation accident and incident rates, underscoring the need for proactive safety measures [2], [3]. Currently, aviation safety heavily relies on reactive anomaly detection, wherein deep learning and data mining techniques are applied to flight data after an event has concluded to interpret causal factors.

A critical gap exists in the literature regarding rigorous computational frameworks that benchmark how early an anomaly can be predicted before catastrophic failure or exceedance occurs. While retrospective analyses are valuable for informing future regulations, operational safety requires forecasting the proactive versus reactive threshold. Furthermore, even when deep learning models achieve high accuracy on aviation datasets, they lack the rigorous explainability required for safety-critical deployment, thereby limiting pilot and operational trust.

To address these limitations, this paper proposes a novel real-time forecasting and diagnostic framework. Our core contributions are twofold:

- **The Real-Time Checkpointing Framework:** We introduce a rigorous methodology for simulating real-time sequence streaming by truncating and zero-padding datasets at 25%, 50%, 75%, 90%, and 100% flight completion marks. This isolates the aerodynamic anomaly precursor phases from the onset and reactive phases, allowing us to benchmark domain-specific baselines against cross-domain architectures and adaptive early classification networks.
- **The Explainability Audit:** We establish an end-to-end interpretation pipeline utilizing Grad-CAM to track dynamic temporal attention, Monte Carlo Dropout to measure epistemic uncertainty and calibration, and segment-level perturbation methods (LIME) to mathematically verify spatial anomaly diagnosis.

II. Related Work

A. Sequence Models in Aviation

A foundational pillar in data-driven aviation safety has been the utilization of large-scale flight repositories, such as NASA's DASHlink. Recent advances have applied deep learning to identify precursors to events like high-speed exceedances. Janakiraman (2018) introduced the Deep Temporal Multiple-Instance Learning (DT-MIL) model, utilizing recurrent architectures [4]. Bleu Laine et al. (2022) advanced this paradigm with the Multi-Head Convolutional Neural Network and Recurrent Neural Network (MHCNN-RNN) [5]. While subsequent extensions of the MHCNN-RNN architecture (e.g., Attn Block 2) were originally made for other domains, we adapt this temporal attention technique, using softmax projections over recurrent outputs, as an advanced aviation baseline.

Beyond aviation, cross-domain state-of-the-art architectures offer alternative inductive biases. Inception Time [6] adapts multi-scale 1D convolutions with parallel, varying receptive fields to capture short and long-term trends simultaneously. Similarly, the Gated Transformer Network (GTN) [7], [9] utilizes dual-pathway self-attention to independently model

step-wise temporal dynamics and channel-wise sensor correlations, fusing them via dynamic soft gating. Incorporating these benchmarks tests whether domain-agnostic pattern matching can surpass specialized aviation architectures.

B. Early Classification & Explainability

Real-time monitoring is fundamentally an early classification problem. The EARLIEST framework [8] integrates an LSTM encoder with a reinforcement learning-based halting policy. By calculating a geometric-like halting distribution at each timestep, it optimizes a composite loss ($L_{total} = L_{cls} + \lambda \frac{\mathbb{E}[t_{halt}]}{T_{max}}$) to autonomously balance classification accuracy against observation latency.

Deployable aviation systems also require rigorous explainability. Metrics like Expected Calibration Error (ECE) via Monte Carlo Dropout [10] ensure fail-safe awareness by explicitly quantifying model uncertainty. Furthermore, while gradient-based methods like Grad-CAM [11] provide rapid heatmaps, perturbation-based methods such as LIME [12] are necessary to mathematically isolate temporal segments, proving the network has learned the correct physical anomaly.

III. Methodology & Experimental Design

A. Dataset and Temporal Framing

This study utilizes multivariate flight data sourced from the NASA DASHlink repository, sampled at 1 Hz. Flight trajectories were spatially resampled over the final 20 nautical miles prior to descending through 1,000 feet above ground level, yielding a fixed-length sequence of 81 time steps per flight. Following dimensionality reduction, the input space was refined to 23 continuous features. The dataset comprises 20,079 naturally balanced flights (48.7% nominal, 51.3% adverse) targeting high-speed exceedances via standard binary cross-entropy.

To evaluate predictive capabilities, we frame the sequence into three phases: Proactive Phase (25%, 50%, 75%) testing precursor prediction; Onset Phase (90%) evaluating instantaneous alerts; and Diagnostic Phase (100%) for retrospective classification.

B. Real-Time Checkpointing Methodology

To simulate real-time data streaming, sequences are truncated at specific checkpoints and zero-padded to restore the 81-timestep tensor dimension, mimicking an active data buffer.

We conducted an ablation study at the 75% boundary utilizing the established baseline, MHCNN-RNN, to evaluate three training strategies: Symmetric (Matched) Splitting, Variable (Mixed) Splitting, and Asymmetric (Full) Splitting.

As Table I details (via 5-fold cross-validation), symmetric splitting is definitively the optimal strategy. The Asymmetric strategy collapses completely at the 75% boundary. This proves models trained on full sequences become fatally dependent on late-stage landing signals and cannot default to earlier precursors when truncated. Failing at the most data-rich proactive checkpoint (75%) guarantees failure at sparser checkpoints. Consequently, symmetric splitting was adopted for all primary benchmark evaluations.

TABLE I
Ablation Study: Training Strategies at the 75% Proactive Boundary (MHCNN-RNN)

Training Strategy	Accuracy	F1-Score	AUC-ROC
Symmetric (75%)	**0.676**	**0.687**	**0.732**
Variable (25–100%)	0.587	0.536	0.642
Asymmetric (100%)	0.505	0.463	0.512

C. Architectures & Training Setup

We establish a comprehensive benchmark across three paradigms. For Aviation Baselines, the MHCNN-RNN utilizes independent 1D convolutional heads (kernel sizes 8, 5, 3) followed by a 20-unit GRU, concluding with Multiple Instance Learning (MIL) max-aggregation. For Cross-Domain SOTA, Inception Time stacks three residual modules containing parallel convolutions (kernels 5, 10, 20) and Global Average Pooling. The Gated Transformer Network (GTN) employs 4-head self-attention ($d = 64$). For Adaptive Early Classification, EARLIEST pairs a 128-unit LSTM encoder with a dense halting network evaluated over a maximum sequence of $T_{max} = 81$.

Models were implemented using TensorFlow/Keras and trained on NVIDIA A100 GPUs via 5-fold cross-validation. Optimization utilized Adam with a uniform batch size of 128. Base learning rates were empirically tuned: 10^{-3} for Inception Time and EARLIEST, 5×10^{-4} for the MHCNN-RNN, and 10^{-4} for GTN. EARLIEST utilized an earliness penalty weight of $\lambda = 0.5$. A `ReduceLROnPlateau` scheduler monitored validation loss (reduction factor 0.5, patience 3, minimum limit 10^{-6}). Regularization included dropout (0.4 for Inception Time, 0.2 for EARLIEST and GTN, and 0.1 for the MHCNN-RNN) alongside L_2 weight decay.

To guarantee every checkpoint model was compared at its peak generalization capacity rather than a potentially overfit final state, we enforced early stopping with a patience of 5 epochs (monitoring validation loss), strictly restoring the best epoch's weights upon termination.

D. Explainable AI (XAI) Audit Pipeline

To ensure deployment viability, optimal models undergo a three-tiered explainability audit:

- **Concept Formation:** We adapt Grad-CAM [11] for multivariate time series to expose spatio-temporal regions of intense focus across all checkpoints.
- **Epistemic Uncertainty:** We adapt Monte Carlo (MC) Dropout [10] utilizing $N = 30$ stochastic forward passes to compare the Expected Calibration Error (ECE) and uncertainty metrics of early vs. late checkpoints.
- **Perturbation Verification:** We employ LIME [12] on the 100% diagnostic models. The sequence is chunked into 10 temporal blocks (≈ 8 time steps each). Systematically masking segments verifies if the network accurately diagnoses late-stage landing anomalies.

TABLE II
5-FOLD CROSS-VALIDATION PERFORMANCE ACROSS REAL-TIME CHECKPOINTS (ACCURACY & AUC-ROC)

Model	25% (Proactive)		50% (Proactive)		75% (Proactive)		90% (Onset)		100% (Diagnostic)	
	Acc	AUC	Acc	AUC	Acc	AUC	Acc	AUC	Acc	AUC
Inception Time	0.603	0.645	0.628	0.674	0.681	0.746	**0.821**	**0.905**	**0.973**	**0.996**
GTN	0.599	0.637	0.625	0.669	0.677	0.742	0.816	0.899	0.963	0.994
Attn Block 2	**0.616**	**0.657**	**0.636**	**0.688**	**0.688**	**0.750**	0.819	0.901	0.942	0.985
MHCNN-RNN	0.615	0.655	0.632	0.684	0.676	0.732	0.810	0.897	0.936	0.982
EARLIEST	0.608	0.639	0.612	0.648	0.660	0.716	0.788	0.867	0.807	0.906
Base GRU	0.594	0.624	0.595	0.630	0.617	0.660	0.696	0.766	0.796	0.880

Note: All models presented demonstrated stable convergence during 5-fold cross validation training.

IV. RESULTS AND DISCUSSION

A. Real-Time Checkpoint Benchmarking

We analyzed classification performance across five operational checkpoints explicitly mapped to three phases: the Proactive Phase (25%, 50%, 75%), the Onset Phase (90%), and the Diagnostic Phase (100%) (Table II). The fundamental observation is a non-linear progression in predictive accuracy.

As demonstrated in Fig. 1, all models demonstrate consistent accuracy gains with marginal improvement throughout the Proactive Phase (+2-5% per step) and dramatic acceleration at the Onset transition (+12-15%). This inflection point validates the framework's foundational premise: the final 25% of the approach contains disproportionately more discriminative signal than the preceding 75% combined.

The data confirms that aerodynamic precursor signals in the Proactive Phase are currently too sparse to trigger high-confidence automated interventions. This limitation suggests that future research must pivot away from expecting high-accuracy binary classification from flight telemetry data alone during early approach windows. Instead, robust proactive anomaly detection will likely require integrating external, multi-modal data streams that precede the physical, aerodynamic manifestation of an exceedance.

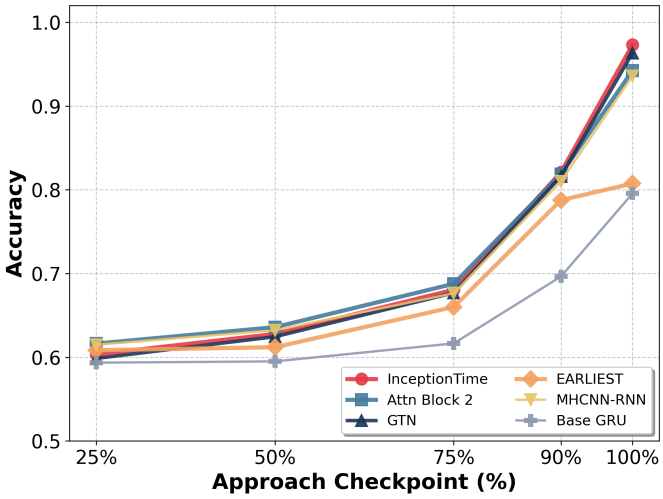

Fig. 1. Classification Accuracy Across Approach Checkpoints. Displays the 90% inflection point and the crossover of model rankings.

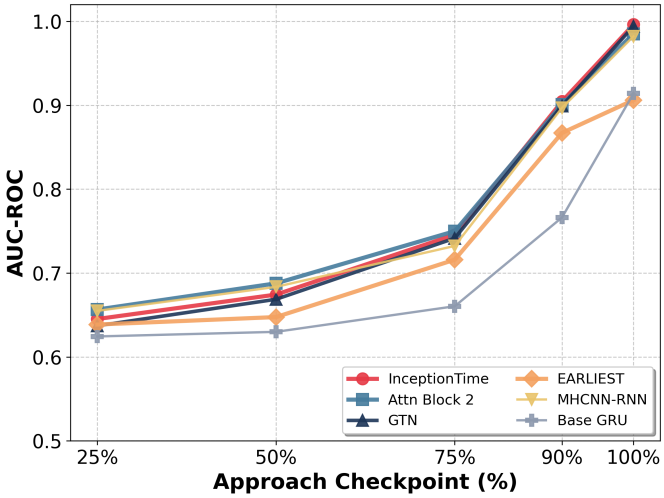

Fig. 2. AUC-ROC Across Approach Checkpoints. Highlights early probability discrimination prior to firm classification boundaries.

B. Model Tier Separation and Crossover

At full-flight evaluation (Diagnostic Phase), models clearly separate into three tiers based on inductive biases: **Tier 1 (96-97%)** multi-scale or self-attention mechanisms (Inception Time, GTN); **Tier 2 (93-94%)** attention-augmented CNN-GRU hybrids (MHCNN-RNN); and **Tier 3 (79-80%)** sequential RNN processing alone (EARLIEST, Base GRU).

However, a critical ranking crossover occurs across the operational phases. The Aviation Baseline (Attn Block 2) consistently leads throughout the early Proactive Phase (25-75%), while Inception Time dominates the late Onset and Diagnostic Phases (90-100%).

This crossover highlights how inductive biases interact with feature availability. Our explainability audit reveals Attn Block 2's RNN-hybrid architecture excels early by tracking subtle, sequential lateral dynamics (e.g., CTAC, LATG). Conversely, Inception Time's multi-scale convolutions perfectly capture the sudden, simultaneous aerodynamic shifts (e.g., FLAP, RALT) defining the Onset Phase. Consequently, monolithic models may be suboptimal; future systems should explore dynamic ensembles that hand off monitoring authority from recurrent tracking to convolutional state-matching at the 75% boundary.

C. Adaptive Halting and Continuous Risk Assessment

Unlike standard architectures, EARLIEST gains only +1.9% accuracy between 90% and 100%. Because its reinforcement learning policy explicitly penalizes observation latency, it willingly trades peak diagnostic accuracy for faster decision-making. Thus, its lower overall accuracy reflects its objective function rather than a structural failure.

Beyond specific architectures, a universal finding is the pronounced disparity between strict accuracy and AUC-ROC across *all* operational checkpoints (Fig. 2). Even at 25%, where accuracy hovers near baseline ($\approx 60\%$), AUC-ROC reaches 0.69, a performance gap that persists through full-flight evaluation. This proves deep learning models capture continuous risk gradients far more effectively than they enforce rigid binary thresholds. Operationally, this confirms that aviation anomalies manifest as gradual state degradations rather than instantaneous failures. Consequently, industry safety systems should pivot from early hard-binary alerts toward continuous, probabilistic risk dashboards, reserving strict alarms exclusively for the highly discriminative Onset Phase.

V. Explainability Analysis

To validate deployability, we conduct an explainability analysis focusing on the top Cross-Domain model (Inception Time) and the MHCNN-RNN aviation baseline variant (Attn Block 2), supplemented by an early-halting analysis of EARLIEST.

A. Gradient Saliency & Feature Convergence

Gradient saliency analysis—filtered to display the 7 most salient flight parameters for visual clarity—reveals a progressive attention migration that closely mirrors the operational phases of an instrument approach. As seen in Fig. 3 and Fig. 4, both architectures exhibit diffuse attention at the 25% checkpoint. Notably, they exhibit distinct inductive biases when data is sparse: Inception Time exhibits diffuse attention across early lateral dynamics (e.g., LATG) with scattered noise across the zero-padded region, while Attn Block 2 produces sharper hotspots on tracking features (CTAC, LATG).

By 50-75%, attention shifts toward engine parameters (N1_1) and aerodynamic state (AOA1). Crucially, at the 90-100% checkpoints, both models independently converge on flap position (FLAP) as the dominant predictor. This architecture-invariant convergence strongly validates FLAP as a genuine causal indicator, eliminating the likelihood of model-specific heuristics.

B. MC Dropout Uncertainty Quantification

MC Dropout uncertainty demonstrates a non-linear maturation of model reliability, as detailed in Table III. For Inception Time, the separation ratio between correctly and incorrectly classified predictions progresses from 0.97x (indistinguishable) at 25% to 14.81x (highly discriminative) at 100%. This massive phase transition precisely coincides with the emergence of configuration-phase features in both of the saliency heatmaps.

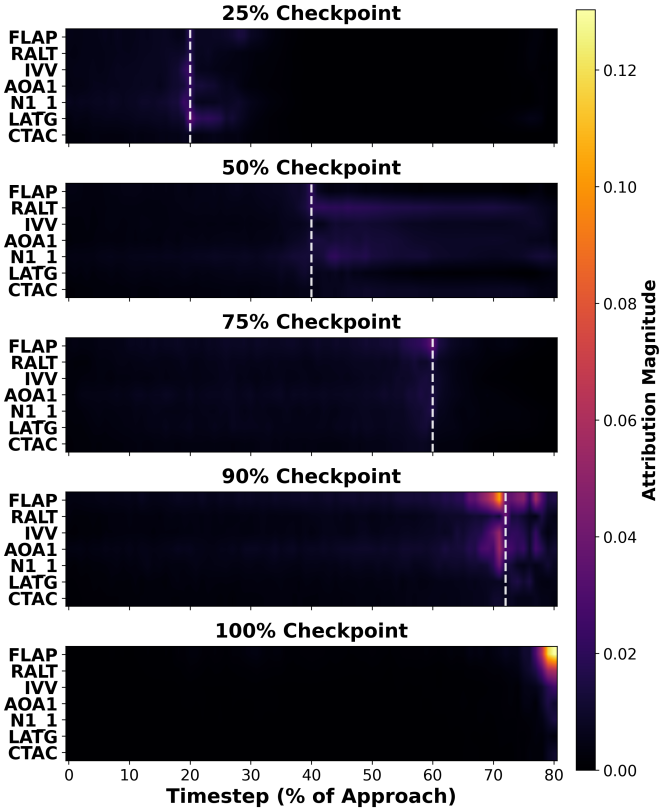

Fig. 3. Inception Time Gradient Saliency Progression. Attention migrates from positional parameters (25%) to extreme concentration on FLAP (100%).

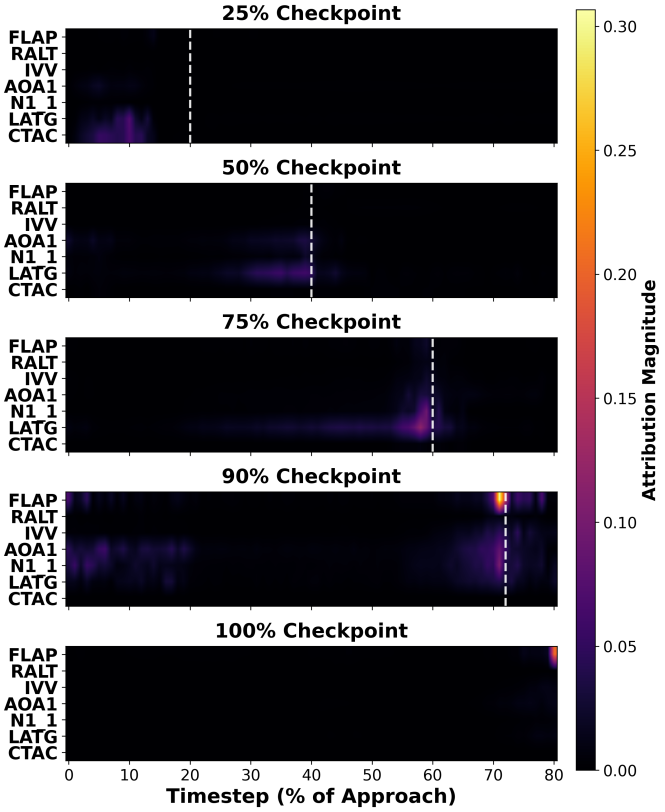

Fig. 4. Attn Block 2 Gradient Saliency Progression. Shows sharper early focus on dynamics, independently converging on FLAP by 100%.

TABLE III
MC DROPOUT UNCERTAINTY METRICS (CORRECT VS. INCORRECT PREDICTIONS)

Model	Phase	ECE	Uncertainty		Separation Ratio
			Correct	Incorrect	
Inception Time	25%	0.0744	0.0016	0.0016	0.97x
	75%	0.0984	0.0029	0.0035	1.23x
	100%	**0.0130**	0.0008	0.0122	**14.81x**
Attn Block 2	25%	**0.0100**	0.0022	0.0023	1.03x
	75%	0.0369	0.0034	0.0043	1.27x
	100%	0.0139	0.0042	0.0289	6.82x

Interestingly, Table III reveals a paradox between calibration and separation. While Attn Block 2 achieves vastly superior Expected Calibration Error (ECE) at the sparse 25% checkpoint (0.0100 vs. Inception Time's 0.0744), Inception Time delivers more than double the uncertainty separation at full flight (14.81x vs. 6.82x). This highlights a critical distinction between population-level calibration and per-prediction reliability. Inception Time's capacity to dramatically spike its uncertainty variance when making an incorrect prediction makes it far superior for triggering automated failsafe alerts.

C. LIME Segment-Level Cross-Validation

Gradient methods provide rapid visual heatmaps but can be mathematically loose. As shown in Fig. 5, LIME independently confirms that the final segment (timesteps 72-81, representing 89-100% of the approach) carries overwhelming importance for Inception Time, yielding an importance score 14x larger than any other flight segment. This cross-method triangulation formally verifies the network's reliance on late-stage anomalies.

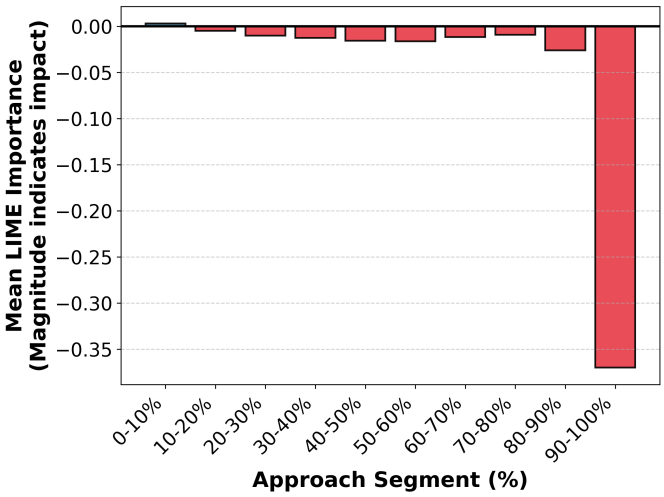

Fig. 5. LIME Segment Importance (Inception Time @ 100%). The final 11% of the approach overwhelmingly dictates the model's prediction.

D. Operationally Grounded Phase Structure

Synthesis of the explainability outputs reveals that the benchmarked architectures naturally learn to segment the approach into operationally meaningful phases without explicit supervision (Table IV). This proves the networks capture genuine aerodynamic causal chains rather than spurious noise. Operationally, this dictates that future diagnostic dashboards should dynamically shift their feature-monitoring priorities to mirror this physics-based landing sequence.

TABLE IV
OPERATIONALLY GROUNDED FEATURE TAXONOMY

Category	Key Features	Active Range
Lateral Dynamics	CTAC, LATG	25–75%
Aerodynamic Config	FLAP, AOA1	75–100%
Engine State	N1_1	50–100%
Altitude Profile	RALT, IVV	90–100%

E. Early Classification Analysis (EARLIEST)

To evaluate adaptive halting, we analyzed the EARLIEST network. Halting behavior is distinctly bimodal (Fig. 6): $\approx 42\%$ of anomalous flights trigger immediate halting (timesteps 0-2), indicating strong initial deviations. Conversely, 71% of nominal flights read the full sequence, adopting a cautious strategy.

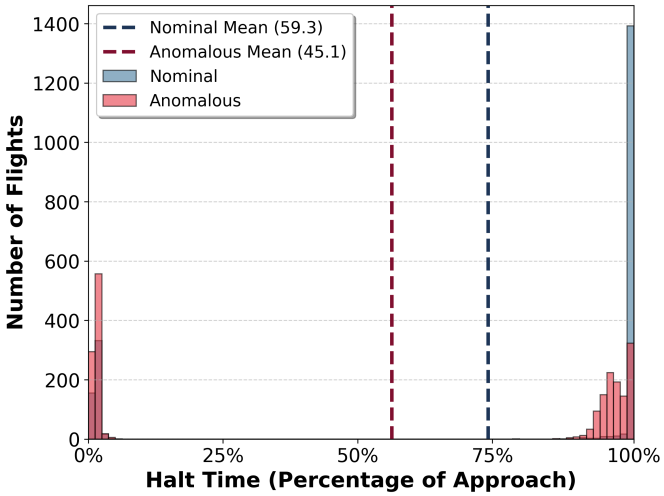

Fig. 6. EARLIEST Halting Time Distribution by Class. Shows a bimodal pattern where a large subset of anomalies is detected almost immediately.

This confirms high-speed exceedances manifest non-uniformly. EARLIEST effectively isolates two anomaly profiles: severe deviations identifiable instantly, and subtle anomalies requiring full-approach context. Furthermore, cross-architecture validation reveals EARLIEST's native feature importance converges on RALT, IVV, and FLAP at the 100% checkpoint. That a fundamentally different RL-based LSTM relies on the exact same features as our convolutional models confirms these parameters are genuine causal indicators rather than model-specific artifacts. Consequently, adaptive halting networks are best deployed as initial "triage" filters. Instant-halt flights can trigger early failsafes, while the remainder are

dynamically handed off to high-accuracy, late-stage convolutional models (e.g., Inception Time) for definitive diagnosis.

VI. Conclusion

This research introduces a dual-contribution framework that bridges the persistent gap between theoretical deep learning performance and practical aviation safety monitoring. By establishing a rigorous real-time checkpointing methodology, we evaluated state-of-the-art sequence models across Proactive, Onset, and Diagnostic flight phases. Our results computationally demonstrate that while early precursor classification is inherently limited by aerodynamic signal sparsity, Onset detection (at the 90% checkpoint) unlocks a highly viable and critical window for instantaneous alerting before a high-speed exceedance fully materializes.

Furthermore, this study proves that complex, black-box architectures can be systematically transformed into transparent, deployment-ready systems. Through a comprehensive explainability audit, we demonstrated that multi-scale architectures like Inception Time not only achieve superior diagnostic accuracy but also exhibit massive epistemic uncertainty separation, making them ideal for triggering automated failsafes. Crucially, cross-method verification utilizing Grad-CAM, LIME, and the EARLIEST network's adaptive halting policy confirmed that these models learn operationally grounded physics—independently converging on late-stage aerodynamic configurations rather than spurious data artifacts.

Ultimately, these findings suggest that future aviation safety systems should pivot from premature binary alarms toward continuous, probabilistic risk dashboards. By leveraging dynamic AI ensembles that adapt to feature availability across flight phases, this framework provides a robust blueprint for deploying trustworthy artificial intelligence in safety-critical environments.

References

[1] Rose et al., "Aviation Safety and Predictive Analytics," *Journal of Aerospace Information Systems*, 2020.

[2] Zhou et al., "Trends in Global Aviation Safety," *IEEE Transactions on Intelligent Transportation Systems*, 2019.

[3] Lu et al., "Machine Learning for Aviation Safety: A Review," *AIAA Aviation Forum*, 2023.

[4] Janakiraman, "Discovering Precursors to Aviation Safety Incidents using Deep Temporal Multiple-Instance Learning," *KDD*, 2018.

[5] Bleu Laine et al., "Intelligent Methodology for the Discovery of Precursors of Adverse Events in Aviation," *IEEE Transactions on Intelligent Transportation Systems*, 2022.

[6] Ismail Fawaz et al., "InceptionTime: Finding AlexNet for time series classification," *Data Mining and Knowledge Discovery*, 2020.

[7] Liu et al., "Gated Transformer Networks for Multivariate Time Series Classification," *arXiv preprint arXiv:2103.14438*, 2021.

[8] Hartvigsen et al., "Adaptive-Halting Policy Network for Early Classification," *KDD*, 2019.

[9] Vaswani et al., "Attention Is All You Need," *NeurIPS*, 2017.

[10] Gal, Y., & Ghahramani, Z., "Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning," *ICML*, 2016.

[11] Selvaraju, R. R., et al., "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization," *ICCV*, 2017.

[12] Ribeiro, M. T., Singh, S., & Guestrin, C., "Why Should I Trust You?': Explaining the Predictions of Any Classifier," *KDD*, 2016.